AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Problem Statement

Development of an AI-based system to identify digital risks and reduce unhealthy algorithm-driven content exposure among young people.

Problem Understanding

- Primary users: Teenagers and young adults aged 16–23, especially college students and frequent social media users. Secondary stakeholders: Parents, teachers, educational institutions, counsellors, and cybersecurity educators.

- Algorithm-driven recommendations repeatedly show similar content to keep users engaged.
  Fragmented safety tools require users to manage privacy settings, screen-time limits, and reporting features separately.
  Rapidly evolving digital threats make it difficult for users to identify harmful content in time.

- Mental well-being: Excessive social media use and social comparison can contribute to stress and anxiety.
  Misinformation: Users may believe and share false information.
  Privacy and security: Personal information may be exposed or misused. Also result in financial losses.
  Parents and institutions: They face difficulties monitoring online safety and educating young people about digital risks.

Existing Gaps & Challenges

- Current digital safety tools are fragmented across different platforms. Users must switch between separate tools for scam detection, fact-checking, privacy protection, and screen-time management.
- Most existing solutions are reactive rather than proactive. They rely on users to recognize harmful content themselves and do not adequately explain how algorithms influence their social media feeds.
- There is no single, unified solution that provides real-time AI-powered digital risk detection, explains algorithmic manipulation, and promotes healthier social media habits in one place.

Our Proposed Solution

- Develop an AI-powered digital safety companion that detects online risks, identifies potentially harmful content, and helps young people understand how social media algorithms influence their behaviour.

- AI Risk Detection: Detects scams, misinformation, and cyberbullying.
  Smart Alerts: Warns users about harmful content.
  Algorithm Awareness: Explains repeated content recommendations.
  Digital Well-being: Promotes healthy screen-time habits.
  Privacy Protection: Helps secure personal information.

- Safer Experience: Early risk warnings.
  Digital Awareness: Recognize online threats.
  Healthy Habits: Reduce excessive scrolling.
  Data Security: Protect personal information.
  User Control: Manage social media use.

Technical Approach

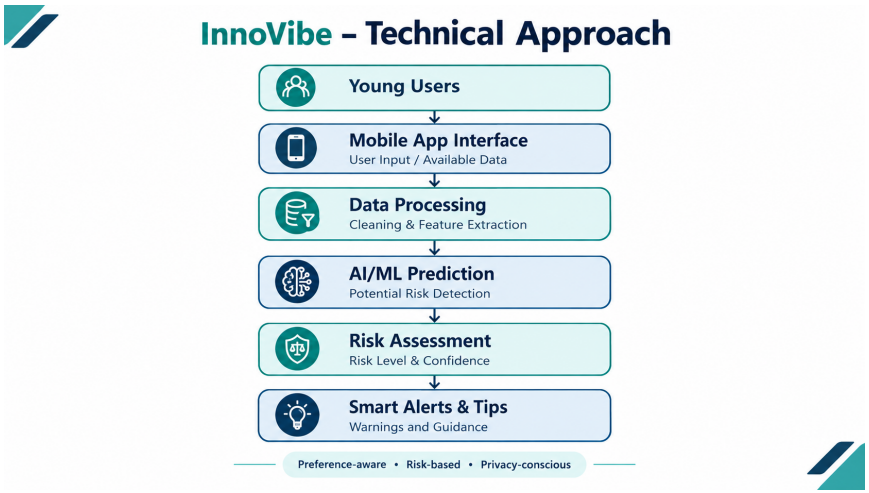


Figure: Architectural diagram

User Input → Data Processing → AI/ML Prediction → Risk Assessment → Smart Alerts and Recommendations

Tech Stack

Frontend: Flutter

Backend: Python(FastAPI)

Database: Firebase Firestore

AI/ML, APIs, or Hardware: Python , Scikit-learn

Innovation & Uniqueness

- What is new or different?: Combines AI-based risk prediction, risk assessment, and safety recommendations in one application.
  Analyses user-provided content instead of relying on direct access to all social media data.
  Provides early warnings about potential digital threats.

- Why is this approach suited to the problem?: Works with limited data access and available information.
  Helps users identify potential risks before taking action.
  Provides practical safety guidance without claiming every prediction is a confirmed fact.

- What is the key differentiator?:
  Predictive rather than purely reactive: Estimates potential risks before users engage with suspicious content.
  Explainable alerts: Explains why content may be risky.
  Privacy-conscious: Uses user-provided and permitted data.
  Unified approach: Combines digital safety awareness and personalised recommendations in one application.

Feasibility & Scalability

- **Feasibility:**
  Resources: Flutter, Python, FastAPI, Firebase, and machine-learning libraries.
  Steps: Build the app → Train/test prediction models → Integrate risk assessment → Display alerts and recommendations.

- **Scalability:** Expand to support more types of digital threats.
  Add more languages and personalised safety recommendations.
  Integrate additional data sources and APIs as they become available.

- **Risks:**
  Limited data: Use user-submitted content and permitted data sources.
  False predictions: Test models and communicate uncertainty.
  Privacy concerns: Minimise data collection and secure stored information.
  API restrictions: Design the app to work without depending entirely on social media APIs.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases

- Use case 1: Scam Detection
  A user submits a suspicious link or message, and the AI predicts potential scam risks and provides safety recommendations.

- Use case 2: Harmful Content Detection
  A user submits a message, and the AI identifies potentially harmful or cyberbullying-related language and suggests appropriate actions.

- Use case 3: Digital Well-being
  A user provides their screen-time or usage data, and the application identifies potentially unhealthy usage patterns and recommends healthier habits.

Impacts and Benefits

- Benefits to Target Users: Helps young people identify potential online threats and make safer digital decisions.
- Social Impact: Promotes digital awareness, online safety, and healthier social media habits.
- Expected Measurable Outcomes: Improved threat awareness, earlier risk identification, and reduced engagement with suspicious content.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target 3.4	Good health and well-being	Promotes healthier digital habits and supports mental well-being
SDG 16	Target 16.10	Peace, Justice and Strong Institutions	Promotes access to reliable information and helps users identify potential online threats and misinformation.

Individual Member Contribution

Member	Contribution or responsibility
Shivani S N	Problem identification
Shruti Ramesh	Idea development
Siddhaarthan S	Presentation design and documentation
Shrisharvesh S V	Analysis of existing solutions

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